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JAWAHARLAL NEHRU UNIVERSITY

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Minor Exam

M.CA. (Winter Semester-2025)

Course Name: Computer Architecture

Maximum Marks: 25

Course Code: CS 405

Total Time: 50 minutes

- Q1. a) What is the difference between RISC and CISC type of Computer Architecture?
b) What is the role of different buses in a Computer system. Explain using 8085 microprocessor.
(3+2)

Q2. (a) What are the major performance parameters that determines the execution time of processor?

b) Explain which performance parameters effects CPI and why?

- (c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. ii) scale up CPI instruction type C to 2.4. Which one is better?

Type	CPI	Frequency
A	1.4	30%
B	2.5	40%
C	2.1	30%

(4+2+4)

Q3. (a) What are the different types of machine cycle in 8085 microprocessor?

- (b) Write an assembly language program to subtract two 4 bit numbers. Determine the total T-state and calculate the total time consumed using 8085 microporocessor. (What is the new memory capacity if the RAM memory is scaled upto from 2^{20} byte to 2^{34} byte?)

(4+4+2)